

Legendre Polynomials and Functions

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September 14, 2026

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Introduction

Legendre's differential equation arises from separation of variables in spherical coordinates. For example, the azimuthally symmetric angular equation is

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \lambda \Theta = 0.$$

Set $x = \cos \theta$ and write $\Theta(\theta) = y(x)$. The chain rule gives every step in the change of variable:

$$\begin{aligned} \frac{d\Theta}{d\theta} &= \frac{dy}{dx} \frac{dx}{d\theta} = -\sin \theta y', \\ \sin \theta \frac{d\Theta}{d\theta} &= -(1 - x^2)y', \\ \frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) &= -\frac{d}{dx} [-(1 - x^2)y'] \\ &= (1 - x^2)y'' - 2xy'. \end{aligned}$$

Introduction (cont.)

Requiring a nonzero solution that remains bounded at both poles $x = \pm 1$ turns this singular Sturm–Liouville problem into an eigenvalue problem. Its eigenvalues are $\lambda = n(n+1)$, $n = 0, 1, 2, \dots$, and hence

$$(1 - x^2)y'' - 2xy' + n(n+1)y = 0. \quad (\text{C.1.1})$$

On an open interval not containing the singular endpoints, two linearly independent solutions are the Legendre functions of the first and second kinds, denoted by $P_n(x)$ and $Q_n(x)$. For nonnegative integer n , P_n is the degree- n solution that is finite at both endpoints, whereas Q_n is singular there.

The polynomial $P_n(x)$ is defined for every real or complex x ; the interval $[-1, 1]$ is its angular and orthogonality domain, not the domain of the polynomial itself. These ordinary Legendre polynomials express the polar-angle dependence of azimuthally symmetric separated solutions of Laplace's and Helmholtz equations; nonzero azimuthal order uses the associated functions introduced below. Exact spherical symmetry would retain only $n = 0$; general spherical-coordinate problems need the higher angular orders.

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Legendre Polynomials

The first few Legendre polynomials are

$$\begin{aligned}P_0(x) &= 1, \\P_1(x) &= x, \\P_2(x) &= \frac{1}{2}(3x^2 - 1), \\P_3(x) &= \frac{1}{2}(5x^3 - 3x), \\P_4(x) &= \frac{1}{8}(35x^4 - 30x^2 + 3), \\P_5(x) &= \frac{1}{8}(63x^5 - 70x^3 + 15x).\end{aligned}\tag{C.2.1}$$

They follow from Rodrigues' formula

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n, \quad n = 0, 1, 2, \dots \tag{C.2.2}$$

That this formula produces a solution of (C.1.1) is verified directly. With $w = (x^2 - 1)^n$ one has $(1 - x^2)w' + 2nxw = 0$; differentiating $n + 1$ times by the Leibniz rule gives

$$\begin{aligned}(1 - x^2)w^{(n+2)} - 2(n+1)xw^{(n+1)} - n(n+1)w^{(n)} \\ + 2nxw^{(n+1)} + 2n(n+1)w^{(n)} = 0,\end{aligned}$$

Legendre Polynomials (cont.)

which collapses to

$$(1 - x^2)w^{(n+2)} - 2xw^{(n+1)} + n(n+1)w^{(n)} = 0,$$

so $w^{(n)}$, and hence P_n , satisfies Legendre's equation. This licenses the use of (C.2.5) for the Rodrigues polynomials in the orthogonality and spectrum arguments below.

Parity, endpoint values, and recurrence

The generating function is

$$g(x, t) = \frac{1}{\sqrt{1 - 2xt + t^2}} = \sum_{n=0}^{\infty} P_n(x) t^n, \quad (C.2.3)$$
$$|t| < 1, \quad -1 \leq x \leq 1.$$

The identification of the expansion coefficients with the Rodrigues polynomials is proved, not assumed. For $|t|$ small enough that $|2xt - t^2| < 1$, the binomial series gives

$$g(x, t) = \sum_{k \geq 0} \frac{(2k)!}{4^k (k!)^2} t^k (2x - t)^k,$$

and collecting the coefficient of t^n from $k = n - j$, $0 \leq j \leq n/2$, yields

$$\sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(-1)^j (2n - 2j)!}{2^n j! (n - j)! (n - 2j)!} x^{n-2j}.$$

Parity, endpoint values, and recurrence (cont.)

Expanding (C.2.2) through $(x^2 - 1)^n = \sum_j \binom{n}{j} (-1)^{n-j} x^{2j}$ produces the same sum, so the coefficient of t^n is exactly P_n . For fixed $x \in [-1, 1]$ the roots of $1 - 2xt + t^2$ are $t = x \pm i\sqrt{1 - x^2}$, both of modulus 1; hence the Taylor series converges for $|t| < 1$. It makes the parity and endpoint values transparent:

$$\begin{aligned} g(-x, t) &= g(x, -t) \\ &= \sum_{n=0}^{\infty} (-1)^n P_n(x) t^n, \end{aligned} \qquad P_n(-x) = (-1)^n P_n(x),$$

$$g(1, t) = (1 - t)^{-1} = \sum_{n=0}^{\infty} t^n, \qquad P_n(1) = 1,$$

$$\begin{aligned} g(-1, t) &= (1 + t)^{-1} \\ &= \sum_{n=0}^{\infty} (-1)^n t^n, \end{aligned} \qquad P_n(-1) = (-1)^n.$$

Parity, endpoint values, and recurrence (cont.)

In particular, $P_n(0) = 0$ for odd n . Differentiating (C.2.3) with respect to t gives

$$(1 - 2xt + t^2) \frac{\partial g}{\partial t} = (x - t)g.$$

Substitution of $g = \sum_{n \geq 0} P_n t^n$ and comparison of the coefficient of t^n yields

$$(n+1)P_{n+1} - 2nxP_n + (n-1)P_{n-1} = xP_n - P_{n-1},$$

and therefore

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x), \quad n \geq 1. \quad (\text{C.2.4})$$

Orthogonality and normalization

Equation (C.1.1) has the self-adjoint form

$$\frac{d}{dx} [(1-x^2)P'_n(x)] + n(n+1)P_n(x) = 0. \quad (\text{C.2.5})$$

Write this equation once for P_n , once for P_m , multiply by P_m and P_n , respectively, and subtract. The product rule then gives

$$\begin{aligned} \frac{d}{dx} \{ (1-x^2) [P_m P'_n - P_n P'_m] \} \\ + [n(n+1) - m(m+1)] P_m P_n = 0. \end{aligned}$$

After integration from -1 to 1 , the boundary term vanishes because P_n, P_m and their derivatives are finite while $1-x^2=0$ at both endpoints. For $m \neq n$, the eigenvalues are distinct, so

$$\int_{-1}^1 P_m(x) P_n(x) dx = 0.$$

Orthogonality and normalization (cont.)

For $m = n$, Rodrigues' formula supplies the norm. Integrating by parts n times (all boundary terms vanish because $(x^2 - 1)^n$ has zeros of order n at $x = \pm 1$) gives

$$\begin{aligned}\int_{-1}^1 P_n^2 dx &= \frac{(-1)^n}{2^n n!} \int_{-1}^1 P_n^{(n)}(x) (x^2 - 1)^n dx \\ &= \frac{(2n)!}{2^{2n} (n!)^2} \int_{-1}^1 (1 - x^2)^n dx.\end{aligned}$$

Here Rodrigues' formula gives $P_n^{(n)} = (2n)!/(2^n n!)$. The beta integral evaluates to

$$\int_{-1}^1 (1 - x^2)^n dx = \frac{2^{2n+1} (n!)^2}{(2n+1)!},$$

so the complete orthogonality and normalization relation, with δ_{mn} the Kronecker delta, is

$$\int_{-1}^1 P_m(x) P_n(x) dx = \frac{2}{2n+1} \delta_{mn}, \quad m, n = 0, 1, 2, \dots \quad (\text{C.2.6})$$

Orthogonality and normalization (cont.)

Consequently, if $f \in L^2[-1, 1]$, its Legendre-series coefficients are obtained by multiplying the expansion by P_m and using (C.2.6):

$$f(x) \sim \sum_{n=0}^{\infty} a_n P_n(x), \quad a_n = \frac{2n+1}{2} \int_{-1}^1 f(x) P_n(x) dx. \quad (\text{C.2.7})$$

Pointwise convergence requires the usual additional regularity conditions; square-integrability guarantees convergence in the L^2 sense.

Orthogonality and normalization

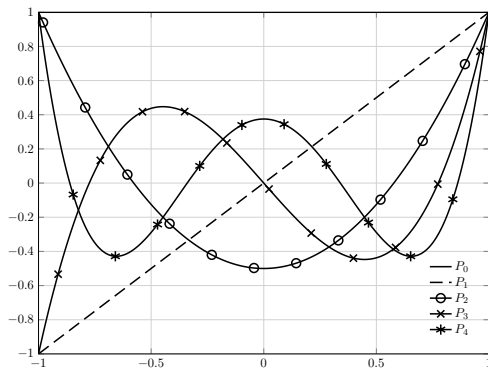


Figure C.1: Legendre polynomials for $n = 0, 1, 2, 3, 4$.

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Associated Legendre Functions

For $-1 \leq x \leq 1$, this book uses the phase-free engineering convention

$$P_n^m(x) = (1 - x^2)^{m/2} \frac{d^m P_n(x)}{dx^m}, \quad 0 \leq m \leq n. \quad (\text{C.3.1})$$

Many mathematical-physics references include the Condon–Shortley factor $(-1)^m$. Thus the convention here and that convention are related by

$$P_{n,\text{here}}^m(x) = (-1)^m P_{n,\text{CS}}^m(x).$$

No differential equation or squared normalization changes under this constant phase. In particular, the convention used here gives the identity employed in Chapter 4,

$$P_n^1(\cos \theta) = \sin \theta P_n'(\cos \theta) = -\frac{d}{d\theta} P_n(\cos \theta).$$

To derive the associated equation without skipping the differentiation step, let $u = d^m P_n / dx^m$. Differentiating (C.1.1) m times and applying the product rule gives

$$(1 - x^2)u'' - 2(m + 1)xu' + [n(n + 1) - m(m + 1)]u = 0. \quad (\text{C.3.2})$$

Associated Legendre Functions (cont.)

Now put $s = 1 - x^2$ and $v = s^{m/2}u = P_n^m$, so that

$$\begin{aligned}u &= s^{-m/2}v, \\u' &= s^{-m/2} \left(v' + \frac{mx}{s}v \right), \\u'' &= s^{-m/2} \left[v'' + \frac{2mx}{s}v' + \frac{m(1 + (m+1)x^2)}{s^2}v \right].\end{aligned}$$

Substitution into (C.3.2) and collection of the v' and v terms gives

$$(1 - x^2)v'' - 2xv' + \left[n(n+1) - \frac{m^2}{1 - x^2} \right] v = 0. \quad (\text{C.3.3})$$

With $x = \cos \theta$, the regular polar factor is $P_n^m(\cos \theta)$. For $m > 0$, multiplication by either real factor $\cos(m\phi)$ or $\sin(m\phi)$, or equivalently use of the complex pair $e^{\pm im\phi}$, supplies the azimuthal dependence. For $m = 0$, the sole nonzero azimuthal factor is the constant 1; $\sin(0\phi)$ is identically zero and is not a second solution.

Associated Legendre Functions (cont.)

For fixed m , equation (C.3.3) has the self-adjoint form

$$\frac{d}{dx} [(1-x^2)v_n'] + \left[n(n+1) - \frac{m^2}{1-x^2} \right] v_n = 0, \quad v_n = P_n^m.$$

Write this equation for v_n and v_l , multiply by v_l and v_n , and subtract:

$$\frac{d}{dx} \{ (1-x^2) [v_l v_n' - v_n v_l'] \} + [n(n+1) - l(l+1)] v_l v_n = 0.$$

After integration on $[-1, 1]$, the boundary term vanishes. For $m = 0$ this is the endpoint argument already used for P_n ; for $m \geq 1$, $v_n = (1-x^2)^{m/2}$ times a polynomial, so each boundary product is $O((1-x^2)^m)$. Thus distinct degrees are orthogonal for fixed m . The norm follows by integrating by parts m times:

$$\begin{aligned} \int_{-1}^1 [P_n^m(x)]^2 dx &= \int_{-1}^1 (1-x^2)^m [P_n^{(m)}(x)]^2 dx \\ &= (-1)^m \int_{-1}^1 P_n(x) \end{aligned}$$

Associated Legendre Functions (cont.)

$$\times \frac{d^m}{dx^m} \left[(1-x^2)^m P_n^{(m)}(x) \right] dx.$$

The required differential identity follows from the self-adjoint form of (C.3.2): with $u = P_n^{(m)}$ that equation reads

$$\frac{d}{dx} \left[(1-x^2)^{m+1} P_n^{(m+1)} \right] = -(n-m)(n+m+1)(1-x^2)^m P_n^{(m)}.$$

Writing $F_m = (1-x^2)^m P_n^{(m)}$, this is $F_m = -F'_{m+1} / [(n-m)(n+m+1)]$; differentiating m times and iterating up to F_n gives

$$F_m^{(m)} = \frac{(-1)^{n-m} (n+m)!}{(n-m)! (2n)!} F_n^{(n)}.$$

Rodrigues' formula makes $P_n^{(n)} = (2n)! / (2^n n!)$ constant, so $F_n = (2n)! / (2^n n!) (1-x^2)^n$ and, by (C.2.2) again, $F_n^{(n)} = (-1)^n (2n)! P_n$. Combining,

$$\frac{d^m}{dx^m} \left[(1-x^2)^m P_n^{(m)}(x) \right] = (-1)^m \frac{(n+m)!}{(n-m)!} P_n(x).$$

Associated Legendre Functions (cont.)

Using (C.2.6), one obtains

$$\int_{-1}^1 P_l^m(x) P_n^m(x) dx = \frac{2}{2n+1} \frac{(n+m)!}{(n-m)!} \delta_{ln}, \quad (\text{C.3.4})$$

for $l, n \geq m \geq 0$. Equivalently, $x = \cos \theta$ and $dx = -\sin \theta d\theta$ give

$$\int_0^\pi [P_n^m(\cos \theta)]^2 \sin \theta d\theta = \frac{2}{2n+1} \frac{(n+m)!}{(n-m)!}. \quad (\text{C.3.5})$$

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Legendre Functions of the Second Kind

On $-1 < x < 1$, the real second solution (more precisely, the Ferrers function of the second kind) is

$$\begin{aligned} Q_n(x) = & \frac{1}{2} P_n(x) \ln \left(\frac{1+x}{1-x} \right) \\ & - \sum_{k=1}^n \frac{1}{k} P_{k-1}(x) P_{n-k}(x), \\ & n = 0, 1, 2, \dots \end{aligned} \tag{C.4.1}$$

where the sum is empty for $n = 0$. For example,

$$\begin{aligned} Q_0(x) &= \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right), \\ Q_1(x) &= \frac{x}{2} \ln \left(\frac{1+x}{1-x} \right) - 1, \\ Q_2(x) &= \frac{3x^2 - 1}{4} \ln \left(\frac{1+x}{1-x} \right) - \frac{3x}{2}. \end{aligned}$$

Legendre Functions of the Second Kind (cont.)

The finite convolution in (C.4.1) is derived as follows rather than verified post hoc. Write $Q_n = \frac{1}{2}P_n L - W_n$ with $L = \ln[(1+x)/(1-x)]$ and W_n a polynomial, and let $\mathcal{L}_n = D[(1-x^2)D] + n(n+1)$. Since $(1-x^2)L' = 2$ is constant, the product rule gives

$$\begin{aligned}\mathcal{L}_n\left[\frac{1}{2}P_n L\right] &= \frac{1}{2}L\mathcal{L}_n[P_n] + (1-x^2)L'P'_n \\ &\quad + \frac{1}{2}P_n[(1-x^2)L']' = 2P'_n,\end{aligned}$$

so Q_n solves (C.1.1) exactly when $\mathcal{L}_n[W_n] = 2P'_n$. For the convolution $W_n = \sum_{k=1}^n \frac{1}{k}P_{k-1}P_{n-k}$, applying the product rule and the Legendre equation to each factor gives

$$\mathcal{L}_n[P_{k-1}P_{n-k}] = 2k(n-k+1)P_{k-1}P_{n-k} + 2(1-x^2)P'_{k-1}P'_{n-k},$$

so the required identity is

$$\sum_{k=1}^n \left[(n-k+1)P_{k-1}P_{n-k} + \frac{1-x^2}{k}P'_{k-1}P'_{n-k} \right] = P'_n. \quad (\text{C.4.2})$$

Legendre Functions of the Second Kind (cont.)

Multiplication by t^n and summation over $n \geq 1$ converts this to a generating-function identity. With $g = \sum_{n \geq 0} P_n t^n$ and $H = \sum_{n \geq 0} P_n t^{n+1}/(n+1)$, so that $H_t = g$ and $H(x, 0) = 0$, the left side of (C.4.2) sums to $tg(tg)' + (1-x^2)H_x g_x$. From $g = (1-2xt+t^2)^{-1/2}$, direct differentiation gives $(tg)' = (1-xt)g^3$ and $g_x = tg^3$; and $\partial_t[(1-x^2)H_x] = (1-x^2)g_x = \partial_t[1 - (1-xt)g]$ with both sides vanishing at $t = 0$, hence $(1-x^2)H_x = 1 - (1-xt)g$. Therefore

$$\begin{aligned}tg(tg)' + (1-x^2)H_x g_x &= t(1-xt)g^4 + tg^3 - t(1-xt)g^4 \\&= tg^3 = g_x,\end{aligned}$$

and the coefficient of t^n in $g_x = \sum_{n \geq 0} P'_n t^n$ is P'_n , proving (C.4.2). Hence Q_n solves (C.1.1); alternatively, these functions follow from Q_0, Q_1 using the same three-term recurrence (C.2.4). To retain the Wronskian sign, divide the differential equation by $1-x^2$ and set $W = P_n Q'_n - P'_n Q_n$. Abel's identity gives

$$W' = \frac{2x}{1-x^2} W \quad \implies \quad W = \frac{C}{1-x^2}.$$

Legendre Functions of the Second Kind (cont.)

As $x \rightarrow 1^-$, $P_n \rightarrow 1$ and the leading logarithmic term in Q_n gives $(1 - x^2)Q'_n \rightarrow 1$, whereas $(1 - x^2)P'_n Q_n \rightarrow 0$; hence $C = 1$. Therefore

$$P_n Q'_n - Q_n P'_n = \frac{1}{1 - x^2},$$

so P_n and Q_n are linearly independent on $(-1, 1)$. The logarithm makes Q_n singular at $x = \pm 1$.

The real Legendre function on either exterior interval $|x| > 1$ uses a different real logarithmic expression,

$$\begin{aligned} Q_n(x) &= \frac{1}{2} P_n(x) \ln \left(\frac{x+1}{x-1} \right) \\ &\quad - \sum_{k=1}^n \frac{1}{k} P_{k-1}(x) P_{n-k}(x), \quad |x| > 1. \end{aligned} \tag{C.4.3}$$

for which the logarithm's ratio is positive. Complex analytic continuation requires a specified branch and must not be obtained by naively inserting $|x| > 1$ into the interior principal-log formula. The exterior $Q_n(\xi)$, with ξ the spheroidal radial coordinate, occurs, for example, as a radial factor in prolate-spheroidal separations. For a spherical exterior problem, by contrast, the radial Laplace factors are $r^{-(n+1)}$, not $Q_n(\cos \theta)$.

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Summary of Important Relations

For $n = 0, 1, 2, \dots$, and $0 \leq m \leq n$, the principal relations are

$$\begin{aligned}\frac{1}{\sqrt{1-2xt+t^2}} &= \sum_{n=0}^{\infty} P_n(x)t^n, \\ |t| < 1, \quad -1 \leq x \leq 1, \\ \int_{-1}^1 P_m(x)P_n(x) dx &= \frac{2}{2n+1} \delta_{mn}, \\ \frac{d}{dx} [(1-x^2)P'_n(x)] + n(n+1)P_n(x) &= 0, \\ \int_0^\pi [P_n^m(\cos \theta)]^2 \sin \theta d\theta \\ &= \frac{2}{2n+1} \frac{(n+m)!}{(n-m)!}.\end{aligned}$$

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Bridge Proofs for Spherical Angular Functions

This section closes the gaps used by Chapter 4: that the regular eigenvalues are exactly $\lambda = n(n+1)$, the admissible associated range $n \geq m$, the angular-Laplacian eigenvalue of the full (polar–azimuthal) mode, and the two angular power identities behind the scattering cross section.

Regular spectrum. Let Θ be nonzero and regular at both poles with $[(1-u^2)\Theta']' + \lambda\Theta = 0$. Subtracting the self-adjoint equation (C.2.5) for P_n cross-multiplied by Θ and P_n and integrating gives

$$(\lambda - n(n+1)) \int_{-1}^1 \Theta P_n du = 0, \quad n = 0, 1, 2, \dots,$$

the boundary term vanishing by the endpoint factor $1-u^2$. Unless $\lambda = n(n+1)$ for some n , every Legendre coefficient of Θ vanishes, and polynomial density (the Weierstrass theorem, taken as an analysis prerequisite) with the orthogonality (C.2.6) forces $\Theta \equiv 0$. Hence the regular spectrum is exactly $\{n(n+1)\}$. This supplies the necessity argument behind series termination in Chapter 4.

Admissible associated range. Writing the associated trial factor as $(1-u^2)^{m/2}V(u)$ with $V(u) = \sum_q b_q u^q$ gives $(1-u^2)V'' - 2(m+1)uV' + [\lambda - m(m+1)]V = 0$, whose

Bridge Proofs for Spherical Angular Functions (cont.)

coefficient recurrence carries the numerator $(q + m)(q + m + 1) - \lambda$. A regular polynomial V therefore requires $\lambda = (N + m)(N + m + 1)$ for the terminating degree N , i.e.

$$n = N + m \geq m.$$

That termination constructs the candidates; that no further regular eigenfunctions exist follows the regular-spectrum argument. For fixed m , cross-multiplying the associated equation for a regular eigenfunction Θ with that for P_n^m and integrating gives

$$(\lambda - n(n + 1)) \int_{-1}^1 \Theta P_n^m du = 0, \quad n = m, m + 1, \dots,$$

the boundary term $(1 - u^2)(\Theta P_n^{m'} - P_n^m \Theta')$ vanishing: for $m = 0$ the endpoint factor $1 - u^2$ multiplies bounded derivatives, and for $m \geq 1$ regularity of $\Theta = (1 - u^2)^{m/2} V$ gives $(1 - u^2)\Theta' = O((1 - u^2)^{m/2}) \rightarrow 0$. Since the derivatives $\{P_n^{(m)}\}_{n \geq m}$ run through a triangular basis of all polynomials, the regular factor $(1 - u^2)^{m/2}\Theta$ is orthogonal to every polynomial, and polynomial density forces $\Theta \equiv 0$ unless $\lambda = n(n + 1)$ for some $n \geq m$. At such an eigenvalue the polynomial V terminates and gives P_n^m ; the second solution is

Bridge Proofs for Spherical Angular Functions (cont.)

singular at a pole. So for each $m \geq 0$ the regular eigenfunctions are exactly the P_n^m with $n \geq m$, consistent with the convention (C.3.1).

Angular-Laplacian eigenvalue of the full mode. With $\nabla_a^2 = \partial_\theta^2 + \cot \theta \partial_\theta + (\sin \theta)^{-2} \partial_\phi^2$ and $\Theta(\theta) = P_n^m(\cos \theta)$, the associated equation in θ reads $(\sin \theta)^{-1} (\sin \theta \Theta')' + [n(n+1) - m^2 / \sin^2 \theta] \Theta = 0$. Acting on the full mode $\Theta(\theta)e^{im\phi}$ (and hence on its real cosine/sine parts),

$$\nabla_a^2 [P_n^m(\cos \theta)e^{im\phi}] = -n(n+1) P_n^m(\cos \theta)e^{im\phi}. \quad (\text{C.6.1})$$

The azimuthal factor supplies the cancelling $-m^2 / \sin^2 \theta$ term. In particular, the polar factor alone is *not* an eigenfunction:

$$\nabla_a^2 P_n^1(\cos \theta) = \left[-n(n+1) + \frac{1}{\sin^2 \theta} \right] P_n^1(\cos \theta).$$

Only the full $P_n^1(\cos \theta) \cos \phi$ (or sine) mode carries the eigenvalue $-n(n+1)$ cancelled in the Chapter 4 interface matching.

Scalar Spherical-Harmonic Completeness

A scalar angular expansion is needed before discussing the vector basis. With the phase-free associated-Legendre convention of these notes, define for $-n \leq m \leq n$

$$Y_{nm}(\theta, \phi) = \left[\frac{2n+1}{4\pi} \frac{(n-|m|)!}{(n+|m|)!} \right]^{1/2} P_n^{|m|}(\cos \theta) e^{im\phi}.$$

The associated-Legendre norm and Fourier orthogonality give $\int Y_{nm}^* Y_{pq} d\Omega = \delta_{np} \delta_{mq}$. To establish completeness rather than only orthogonality, suppose $f \in L^2(S^2)$ is orthogonal to every Y_{nm} . Put $u = \cos \theta$ and take the azimuthal Fourier coefficient

$$f_m(u) = \frac{1}{2\pi} \int_0^{2\pi} f(u, \phi) e^{-im\phi} d\phi.$$

For each integer m , orthogonality gives $\int_{-1}^1 f_m(u) P_n^{|m|}(u) du = 0$ for all $n \geq |m|$. Since $P_n^{|m|} = (1-u^2)^{|m|/2} P_n^{(|m|)}$, and $\{P_n^{(|m|)} : n \geq |m|\}$ is a triangular basis of all polynomials, the function $g_m = (1-u^2)^{|m|/2} f_m$ is orthogonal to every polynomial. Polynomial density gives $g_m = 0$ almost everywhere. Its prefactor is nonzero on $(-1, 1)$, so $f_m = 0$ almost everywhere. Fourier completeness then gives $f = 0$. Hence the Y_{nm} form a complete orthonormal basis, not merely an orthogonal family.

Scalar Spherical-Harmonic Completeness (cont.)

The smooth-field uses of this expansion also require differentiation. For smooth f , integration by parts with the angular Laplacian gives, for $n \geq 1$ and every positive integer p ,

$$|\langle Y_{nm}, f \rangle| \leq [n(n+1)]^{-p} \|(\nabla_a^2)^p f\|_{L^2}.$$

The required polynomial bounds on basis derivatives follow without an asymptotic formula. Rotations preserve the $(2n+1)$ -dimensional eigenspace, so the basis-independent sum $\sum_m |Y_{nm}(\Omega)|^2$ is constant. Integrating it over the sphere fixes that constant as $(2n+1)/(4\pi)$. For the rotation derivatives $D_j = (\hat{e}_j \times \hat{r}) \cdot \vec{\nabla}_a$, integration by parts and $\sum_j |D_j f|^2 = |\vec{\nabla}_a f|^2$ give

$$\sum_j \|D_j f\|_{L^2}^2 = n(n+1) \|f\|_{L^2}^2$$

within this eigenspace. Each D_j preserves it. Repetition and the pointwise bound from the constant sum therefore bound any fixed number q of rotation derivatives by a constant times $\sqrt{2n+1} [n(n+1)]^{q/2} \|f\|_{L^2}$. On regular coordinate patches these derivatives span the tangential derivatives. Combined with the arbitrarily large p above,

Scalar Spherical-Harmonic Completeness (cont.)

these estimates give uniform convergence of each fixed differentiated expansion of a smooth field.

Angular power identities. Put $U_n(\theta) = P_n^1(\cos \theta)$, $\pi_n = U_n / \sin \theta$, $\tau_n = U'_n$. From (C.3.4) with $m = 1$,

$$\int_0^\pi U_n U_\ell \sin \theta \, d\theta = \frac{2n(n+1)}{2n+1} \delta_{n\ell}.$$

Since $U_n = O(\sin \theta)$ at the poles, integration by parts with the $m = 1$ angular equation gives the diagonal identity. Explicitly, that equation reads $(\sin \theta U'_\ell)' = -[\ell(\ell+1) - \sin^{-2} \theta] U_\ell \sin \theta$, so the derivative half of the integral becomes

$$\begin{aligned} \int_0^\pi U'_n U'_\ell \sin \theta \, d\theta &= [\sin \theta U_n U'_\ell]_0^\pi - \int_0^\pi U_n (\sin \theta U'_\ell)' \, d\theta \\ &= \ell(\ell+1) \int_0^\pi U_n U_\ell \sin \theta \, d\theta - \int_0^\pi \frac{U_n U_\ell}{\sin \theta} \, d\theta, \end{aligned}$$

Scalar Spherical-Harmonic Completeness (cont.)

the boundary term vanishing by $U_n = O(\sin \theta)$. Adding the $\pi_n \pi_\ell$ part cancels the last integral, leaving

$$\begin{aligned} \int_0^\pi (\pi_n \pi_\ell + \tau_n \tau_\ell) \sin \theta \, d\theta &= \ell(\ell + 1) \int_0^\pi U_n U_\ell \sin \theta \, d\theta \\ &= \frac{2n^2(n+1)^2}{2n+1} \delta_{n\ell}, \end{aligned} \tag{C.6.2}$$

while the mixed combination needs no spectrum at all:

$$\int_0^\pi (\pi_n \tau_\ell + \tau_n \pi_\ell) \sin \theta \, d\theta = [U_n U_\ell]_0^\pi = 0. \tag{C.6.3}$$

With the azimuthal integrals $\int_0^{2\pi} \cos^2 \phi \, d\phi = \int_0^{2\pi} \sin^2 \phi \, d\phi = \pi$, the diagonal weight against the $[(2n+1)/(n(n+1))]^2$ amplitude prefactors reduces to $2(2n+1)$, yielding the modal scattering sum used in Chapter 4 with all cross terms eliminated.